Task 1: Create a CPU utilization alert

Use AWS CloudWatch for this practical.
1. Open CloudWatch → Alarms → Create alarm.
2. Select your EC2 instance's CPUUtilization metric.
3. Configure the threshold as Greater than 70%.
4. Set the evaluation period to five minutes.
5. Add an SNS notification action and select your confirmed email subscription.
6. Name the alarm High-CPU-Alert and create it.

Task 2: Set up a storage usage alert

A practical approach is:
1. Open CloudWatch → Alarms → Create alarm.
2. Select an appropriate S3 storage metric, such as BucketSizeBytes, if available.
3. Configure the bucket and storage type dimensions.
4. Set a threshold in bytes based on your chosen storage limit.
5. Attach an SNS notification action

Task 3: Configure email notifications

1. Open Amazon SNS in the AWS Console.
2. Create a standard topic called Cloud-Alert-Notifications.
3. Create an email subscription using your Gmail or Outlook address.
4. Open your email and confirm the subscription.
5. Return to CloudWatch and attach this SNS topic to your CPU alarm.
6. Verify that the alarm action is enabled.

Task 4: Configure SMS notifications

1. Open the Amazon SNS console.

2. Review the SMS messaging configuration and available delivery options for your account and region.
3. Configure an SMS destination using a number you own, following AWS verification and account requirements.
4. Review applicable SMS quotas, spending limits and pricing before enabling delivery.

Task 5: Test the alerts

- Generate CPU load on your own test EC2 instance using the safe lab procedure from Assignment 2.
- For storage, upload test files to your own bucket or use a test volume to increase usage.
- Check the CloudWatch alarm state and notification delivery.